Band-1: Gauss Circle — Error-Term Exponent Probe (collapse 1.12)

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Band-1: Gauss circle error-term probe (H-1.12)

Honesty stamp. Not a full close. Increment / ledger only. **No** new proven exponent. **No** Clay stamp. **No** ToE. **GATE-04 LOCKED.** Finite box discrepancy is **illustration only**.

This short probe **adopts** the shipped census `2026-09-11_Band1_Computational_Experiment_gauss_circle` (no recompute).

Normalization

$$N(R) = \#\{(x,y) \in \mathbb{Z}^2 : x^2 + y^2 \le R^2\}, \qquad P(R) = N(R) - \pi R^2 = O(R^{\theta+\varepsilon}).$$

(Equivalently, with $X = R^2$, many papers write $R(X) = O(X^{\alpha+\varepsilon})$; then $\theta = 2\alpha$.)

Literature ledger (best published θ)

Year	θ (radius form)	Authors	Status
2003	$131/208 \approx 0.629808$	Huxley, *Proc. LMS* (3) **87**	previous record
2023	$2\alpha^* \approx 0.628966351948$	**Li–Yang**, arXiv:2308.14859	**best known**
—	$1/2$	Hardy	**OPEN** (conjecture)

- **Primary cite.** Xiaochun Li, Xuerui Yang, *An improvement on Gauss's Circle Problem and Dirichlet's Divisor Problem*, arXiv:2308.14859 (v2, 2023). Prove $R(X) = O_\varepsilon(X^{\alpha^*+\varepsilon})$ with $\alpha^* \approx 0.3144831759741$; equivalently $\theta \le 2\alpha^*$ here.
- **Previous.** M. N. Huxley, *Exponential sums and lattice points III*, Proc. London Math. Soc. (3) **87** (2003): $\theta \le 131/208$ (X-form 131/416).
- **2024–2026 check.** No accepted published improvement of the Li–Yang **upper** bound located (2026-09-11). Sources still citing Li–Yang / Huxley as the standing upper records include Tao expdb §21, arXiv:2609.04725 (numerical Hardy evidence), and Berndt–Moree *Expo. Math.* (2025) survey note. Recent Ω-result refinements (e.g. arXiv:2605.21476) strengthen **lower** bounds only. A 2026 Zenodo claim of optimal $O(R^{1/2})$ is **not** elevated.

Hardy Ω-lower: $\theta \ge 1/2$. Method barrier near radius-form $5/8 = 0.625$ remains.

Box discrepancy (adopted; illustration)

Integer grid $R = 1, \ldots, 10^4$ from Computational_Experiment lab `/workspace/experiments/band1/gauss_circle/` → `results.json`.

Quantity	Value
$R_{\max}$	10^4
$\max \|P(R)\|$	≈ 725.022308 at $R = 8399$
Descriptive θ_{fit} (running-max log–log)	≈ 0.593 (**not a proof**)

Claim boundary. These numbers do **not** beat Huxley or Li–Yang. They do **not** prove Hardy's $\theta = 1/2$.

Paths

- Lab (mission-control): `/workspace/mission-control/experiments/band1/gauss_circle/`
- Full experiment + census: `/workspace/experiments/band1/gauss_circle/`
- Full paper: `papers/2026-09-11_Band1_Computational_Experiment_gauss_circle.{md,pdf}`
- This probe: `papers/2026-09-11_Band1_Gauss_circle_error_probe.{md,pdf}`

Verdict

Best cited θ: Li–Yang $2\alpha^* \approx 0.628966351948$. **Box max discrepancy:** ≈ 725.02 at $R = 8399$. Hardy conjecture: **OPEN**. GATE-04 LOCKED.